

**Relationship between NLP Applications → NLP Pipeline Stages →
Common NLP Tools.**

NLP Application	Real-world Example	NLP Pipeline Stages Involved	Common NLP Tools/Frameworks
Spam Email Detection	Classify emails as Spam/Not Spam	Preprocessing → Morphological → Semantic → Application (Classification)	NLTK, spaCy, Hugging Face
Sentiment Analysis	Customer reviews, social media analysis	Preprocessing → Morphological → Syntactic → Semantic → Application	NLTK, spaCy, Hugging Face
Text Classification	News categorization, document classification	Preprocessing → Morphological → Semantic → Application	NLTK, spaCy, Hugging Face
Machine Translation	English to Hindi translation	Preprocessing → Morphological → Syntactic → Semantic → Pragmatic → Application	Hugging Face, OpenNMT, IndicTrans
Question Answering (QA)	AI assistants, ChatGPT-like systems	Preprocessing → Semantic → Pragmatic → Application	Hugging Face, spaCy
Chatbots / Virtual Assistants	Customer support chatbot	Preprocessing → Semantic → Pragmatic → Application	Rasa, spaCy, Hugging Face
Named Entity Recognition (NER)	Extract names, locations, organizations	Preprocessing → Morphological → Semantic	spaCy, Hugging Face
Information Extraction	Resume parsing, invoice processing	Preprocessing → Morphological → Syntactic → Semantic	spaCy, NLTK
Text Summarization	Summarizing legal or news documents	Preprocessing → Syntactic → Semantic → Application	Hugging Face (BART, T5)
Speech-to-Text	Voice assistants	Preprocessing → Morphological → Semantic → Application	Whisper, Hugging Face

NLP Application	Real-world Example	NLP Pipeline Stages Involved	Common NLP Tools/Frameworks
Part-of-Speech (POS) Tagging	Grammar checking	Preprocessing → Morphological → Syntactic	NLTK, spaCy
Grammar Checking	Grammarly	Preprocessing → Morphological → Syntactic → Semantic	LanguageTool, spaCy
Topic Modeling	Discover topics in documents	Preprocessing → Semantic → Application	Gensim, NLTK
Keyword Extraction	Search engines, SEO	Preprocessing → Morphological → Semantic	spaCy, YAKE, KeyBERT
Document Clustering	Organizing research papers	Preprocessing → Semantic → Application	Scikit-learn, Gensim
Fake News Detection	Social media verification	Preprocessing → Semantic → Pragmatic → Application	Hugging Face, spaCy
Intent Detection	Banking or travel chatbot	Preprocessing → Semantic → Pragmatic	Rasa, Hugging Face
Recommendation Systems (Review-based)	Product recommendations	Preprocessing → Semantic → Sentiment Analysis → Application	Hugging Face, spaCy
Language Identification	Detect English/Hindi/Marathi	Preprocessing → Morphological → Application	fastText, langdetect
Plagiarism Detection	Academic integrity systems	Preprocessing → Semantic → Application	NLTK, spaCy

Mapping of Pipeline Stages

Pipeline Stage	Purpose
Text Preprocessing	Tokenization, stop-word removal, punctuation removal, normalization
Morphological Analysis	Stemming, lemmatization, word formation analysis
Syntactic Analysis	Parsing, grammatical structure, POS tagging
Semantic Analysis	Understanding word and sentence meaning
Pragmatic Analysis	Understanding context, user intent, discourse
Application-Level Analysis	Final NLP task such as classification, translation, summarization, chatbot response

Quick Reference for Students

Application	Most Important Pipeline Stage	Preferred Tool
Spam Detection	Semantic Analysis	Hugging Face
Sentiment Analysis	Semantic Analysis	Hugging Face
Chatbot	Pragmatic Analysis	Rasa / Hugging Face
Machine Translation	Semantic + Pragmatic	Hugging Face
Resume Parsing	Semantic Analysis	spaCy
Named Entity Recognition	Semantic Analysis	spaCy
POS Tagging	Morphological + Syntactic	NLTK / spaCy
Text Summarization	Semantic Analysis	Hugging Face
Question Answering	Pragmatic Analysis	Hugging Face
Fake News Detection	Semantic + Pragmatic	Hugging Face